Blockchain-Powered AI Healthcare Insights

1. Executive Summary

The Blockchain-Powered AI Healthcare Insights platform is designed to create a secure, tamper-proof, and intelligent healthcare ecosystem. By integrating Multi-Chain Blockchain, IPFS-based decentralized storage, AI powered by Ollama, and a Streamlit-based application layer, the system ensures data integrity, privacy, transparency, and predictive healthcare analytics.

2. Problem Statement

Current healthcare systems face the following major challenges:

• Fragmented patient records across institutions

• High risk of data breaches and manipulation

• Lack of patient ownership and access transparency

• Limited trust between hospitals, labs, and insurers

• Poor adoption of AI due to unreliable data sources

The core problem is designing a secure, decentralized, and AI-enabled healthcare data platform that guarantees integrity, privacy, and intelligent insights.

3. Proposed Solution

Our solution integrates Multi-Chain Blockchain for decentralized trust, IPFS for secure off-chain storage, Ollama for AI-based healthcare insights, and Streamlit for an interactive web interface.

4. System Architecture

The system consists of the following layers:

Blockchain Layer (Multi-Chain):

• Stores cryptographic hashes of medical records

• Maintains immutable audit logs

• Enables access control via smart contracts

Storage Layer (IPFS):

• Stores encrypted medical records off-chain

• Ensures distributed, content-addressed storage

• Links file hashes to blockchain entries

AI Layer (Ollama):

• Processes medical reports and generates insights

• Performs summarization and risk analysis

• Enables natural language interaction with healthcare data

Application Layer (Streamlit):

• Role-based login (Patient / Doctor / Admin)

• Dashboard for uploading and retrieving records

• Displays AI-generated healthcare insights

5. Technology Stack

Blockchain:

• Multi-Chain Blockchain Network

• Smart Contract-based access logic

• Cryptographic hashing (SHA-256)

Decentralized Storage:

• IPFS (InterPlanetary File System)

Artificial Intelligence:

• Ollama (Local LLM execution)

• Python-based AI processing pipeline

Frontend & Application:

• Streamlit Web Framework

• Python

Security Mechanisms:

• AES Encryption for records

• Wallet-based identity mapping

• Role-based access control

6. Data Flow

1. User uploads a medical report via Streamlit interface.

2. The report is encrypted and stored in IPFS.

3. The IPFS hash is stored on Multi-Chain blockchain.

4. Ollama processes report data and generates insights.

5. Insights are displayed in the dashboard securely.

7. Security & Privacy

• Immutable blockchain records prevent tampering.

• IPFS ensures distributed file storage.

• Encryption protects sensitive medical information.

• Access permissions are controlled by blockchain logic.

8. Advantages

• Tamper-proof healthcare records

• Decentralized patient data ownership

• Transparent auditing

• AI-powered preventive healthcare insights

• Reduced fraud and improved trust

9. Future Enhancements

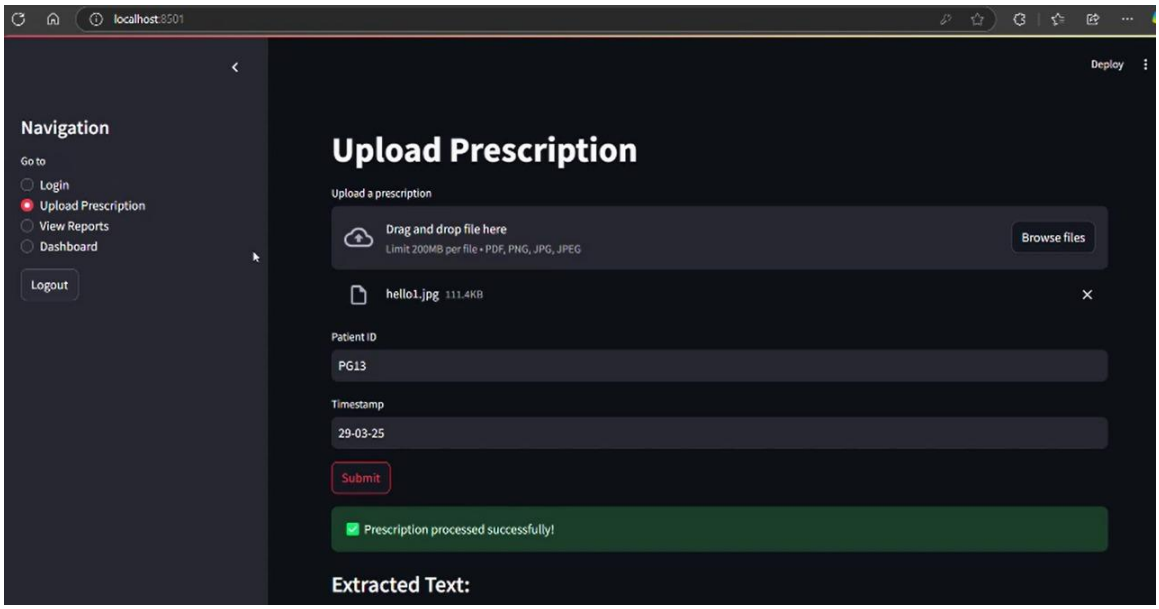
• Federated learning for privacy-preserving AI

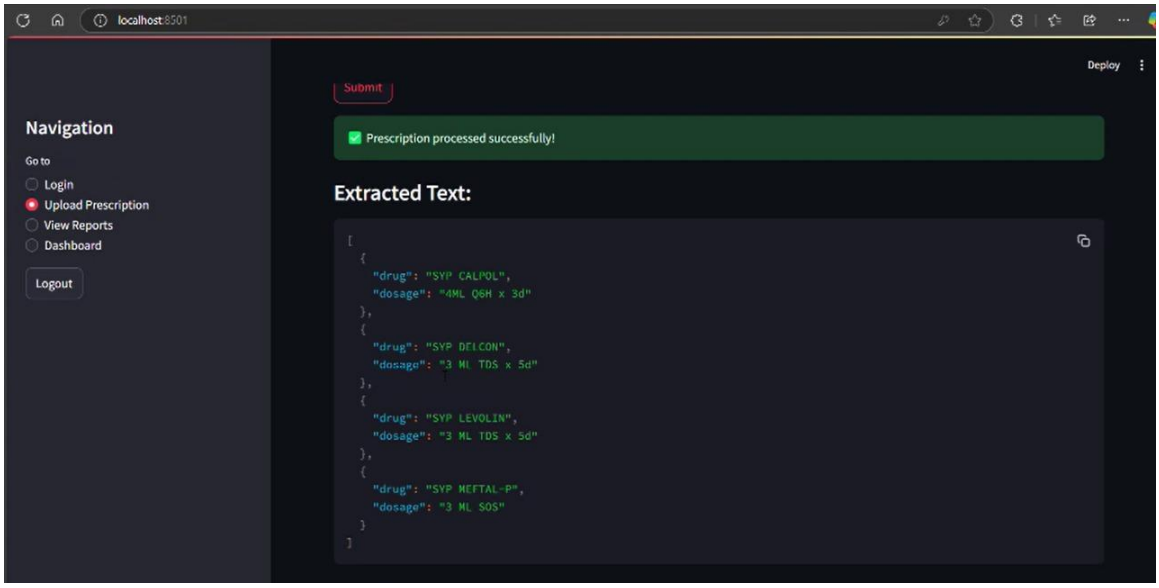
• Zero-knowledge proof-based access validation

• Integration with IoT wearable devices

• Scalable multi-node blockchain deployment

10. Screenshots Outputs





localhost:8501

Deploy

Navigation

Go to

- Login
- Upload Prescription
- View Reports
- Dashboard

Logout

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DDI Analysis:

Based on the prescription drug list provided, there are several potential drug interactions that could be of concern. Here is a summary of the interactions:

1. Severe interaction: Clindamycin (Drug 1) can cause a decrease in the absorption of Kaolin (Drug 2), resulting in a reduced serum concentration and potentially a decrease in efficacy.
2. Moderate interaction: Lincomycin (Drug 1) can also cause a decrease in the absorption of Kaolin (Drug 2).
3. Severe interaction: Tenofovir disoproxil (Drug 1) may decrease the excretion rate of Cidofovir (Drug 2), which could result in a higher serum level.
4. Moderate interaction: Tenofovir disoproxil (Drug 1) can increase the serum concentration of Ganciclovir (Drug 2) and Valganciclovir (Drug 2) when used together.
5. Mild interaction: Ulipristal (Drug 1) can decrease the therapeutic efficacy of Nomegestrol (Drug 2) when used together.
6. Mild interaction: Ganciclovir (Drug 1) can increase the risk or severity of adverse effects when used in combination with Entecavir (Drug 2).
7. Mild interaction: Ascorbic acid (Drug 1) can increase the excretion rate of Deferoxamine (Drug 2), which could result in a lower serum level.

Based on these interactions, here are some alternative medications that may be suitable for the patient:

1. For the Kaolin interaction, an alternative antacid such as calcium carbonate or magnesium hydroxide could be considered.
2. For the Tenofovir disoproxil interaction, a different nucleoside reverse transcriptase inhibitor such as abacavir or lamivudine could be used instead.
3. For the Ganciclovir interaction, an alternative antiretroviral drug such as raltegravir or elvitegravir could be considered.
4. For the Ulipristal interaction, an alternative progesterone-based medication such as norethindrone or medroxyprogesterone could be used